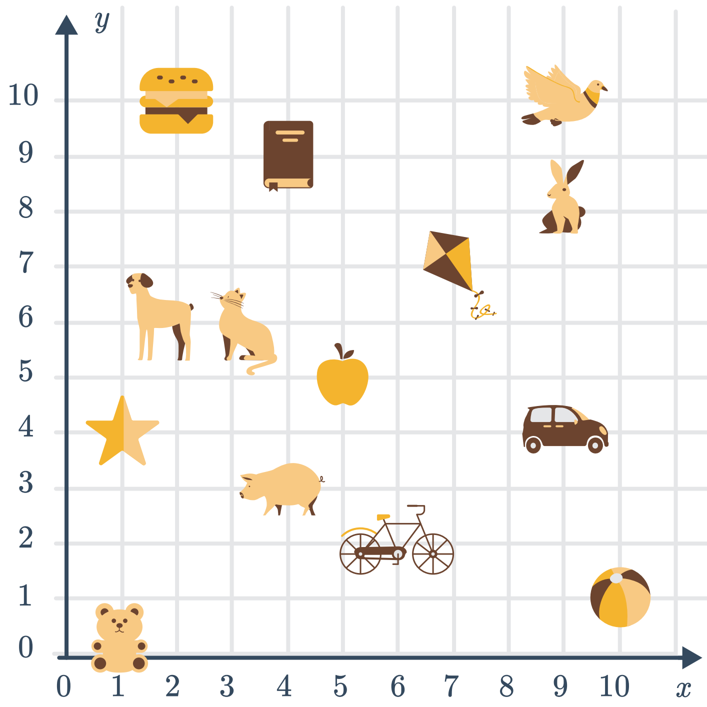
Number plane

1 Consider the number plane shown:



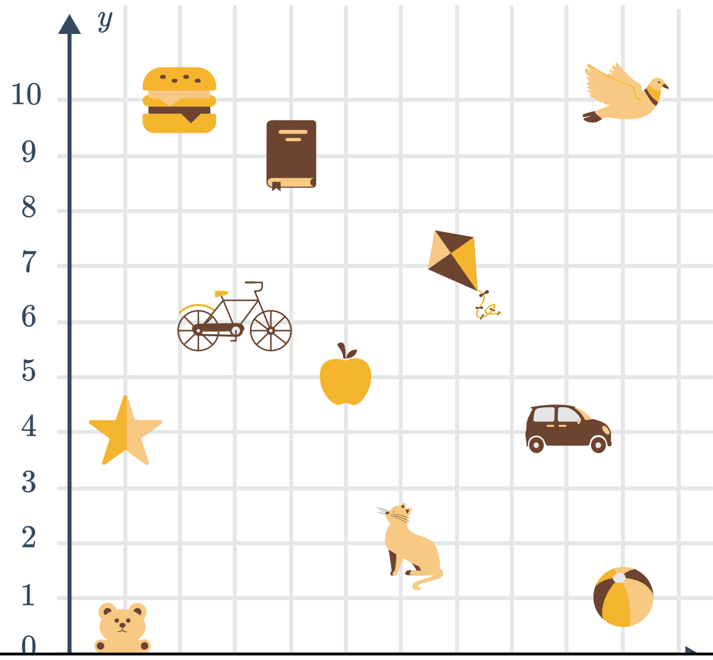
a Identify the object that has the following coordinates:

i $(6, 2)$ ii $(4, 3)$ iii $(2, 6)$ iv $(9, 10)$

b Write the coordinates of the following objects:

i Rabbit ii Cat iii Apple iv Book

2 Consider the number plane shown:



a Identify the object that has the following coordinates:

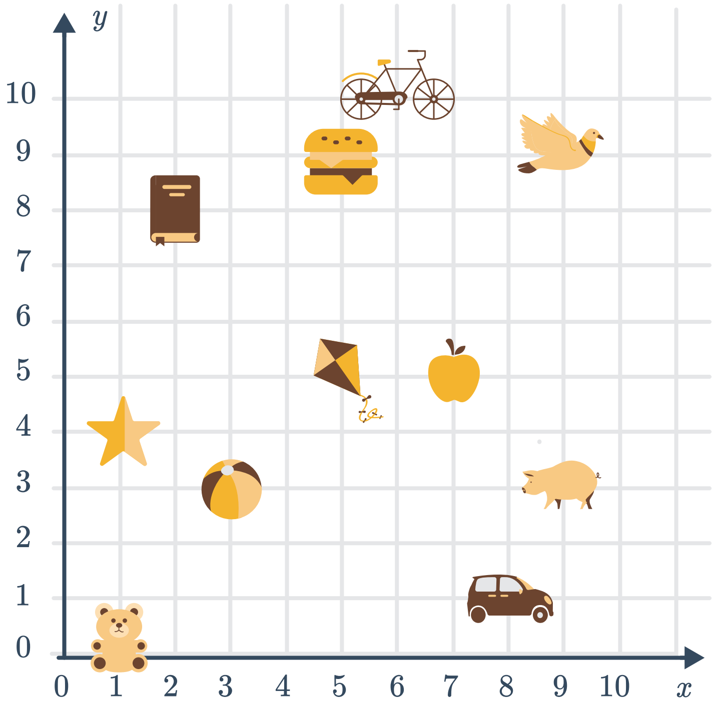
i $(1, 4)$ **ii** $(3, 6)$ **iii** $(5, 9)$ **iv** $(10, 1)$

b Write the coordinates of the following objects:

i Kite **ii** Car **iii** Teddy bear **iv** Bird

3 Consider the number plane shown:

a Identify the object that has the following coordinates:

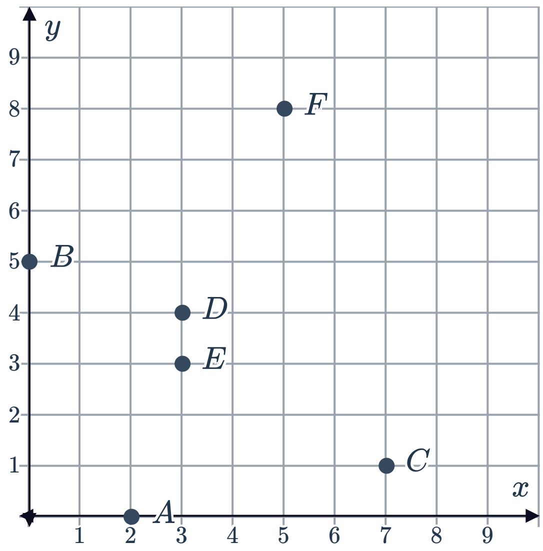


i $(8, 1)$ **ii** $(1, 0)$ **iii** $(9, 3)$ **iv** $(6, 10)$

b Write the coordinates of the following objects:

i Star **ii** Burger **iii** Kite **iv** Beach ball

4 Write down the coordinates of the following points:



a A

b B

c C

d D

e E

f F

5 Plot the following points on a number plane:

a $A\,(3,0)$

b $B\,(0,8)$

c $C\,(5,6)$

d $D\,(6,2)$

e $E\,(7,7)$

f $F\,(0,0)$

g $G\,(3,5)$

h $H\,(7,2)$

i $I\,(0,6)$

j $J\,(9,3)$

k $K\,(7,6)$

l $L\,(5,9)$

m $M\,(8,8)$

n $N\,(8,2)$

o $O\,(9,0)$

p $P\,(0,2)$

q $Q\,(8,6)$

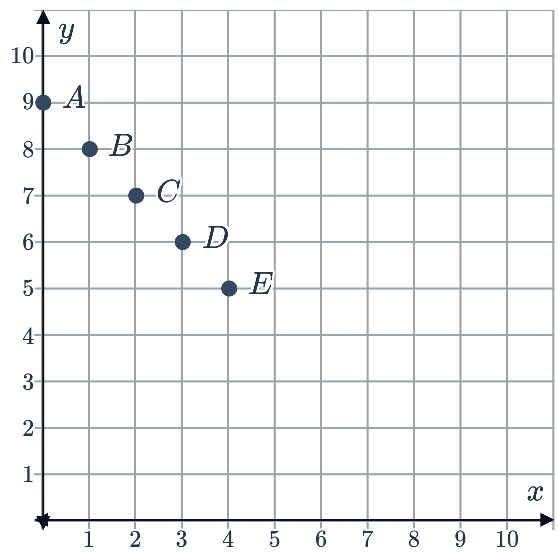
6 Consider the three points: $A\,(3,5)$, $B\,(2,8)$ and $C\,(4,2)$

a Plot these points on a number plane.

b Which point is closest to the x-axis?

c Which point is closest to the y-axis?

7 Consider the following number plane:

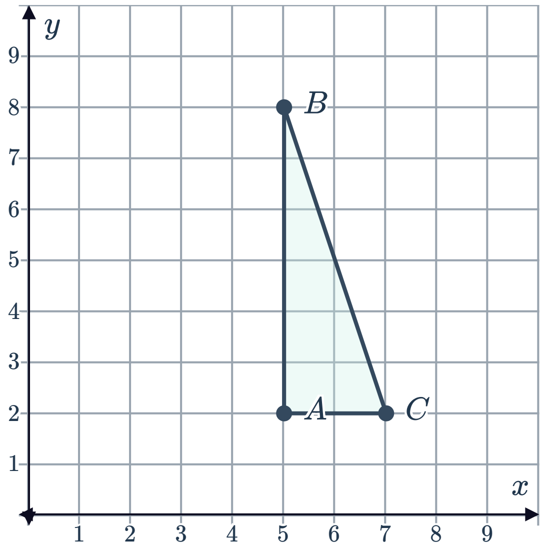


a For all of the plotted points, is the x or y-coordinate larger?

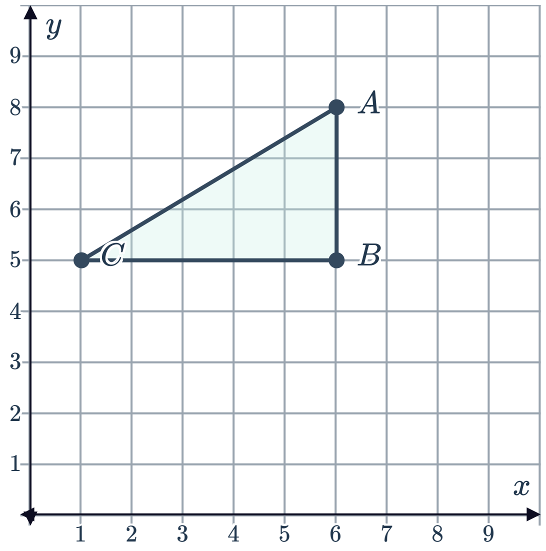
b What does the x-coordinate and y-coordinate of each point add up to?

8 Write down the coordinates of the vertice[redacted]ach of the following shapes:

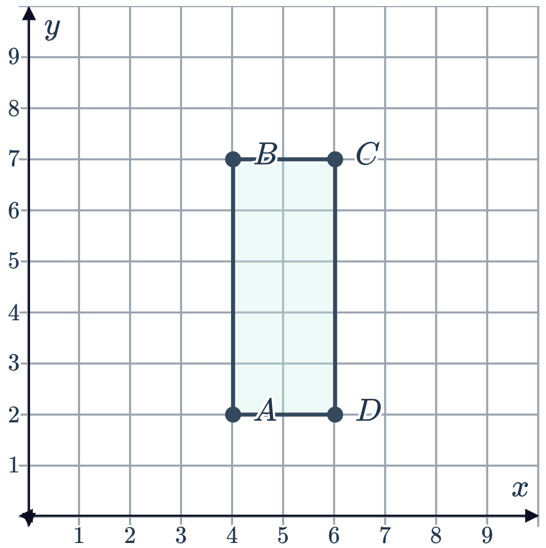
a



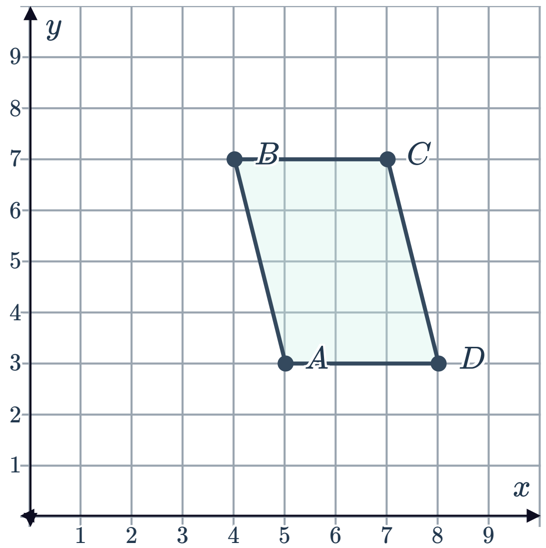
b



c



d



9 Describe in words, how to get to the point $(9, 4)$ from the origin on the number plane.

10 Find the coordinates of the following points:

- **a** The point six units to the right of the origin.
- **b** The point four units above the origin.
- **c** The point five units to the right of $(9, 6)$.
- **d** The point seven units to the left of $(7, 8)$.
- **e** The point one unit to the left and eight units below the point $(3, 10)$.
- **f** The point three units to the right and two units above the point $(2, 5)$.

11 Which of the following points is furthest from the origin?

A $(0, 3)$ **B** $(0, 10)$ **C** $(6, 0)$ **D** $(9, 0)$

12 Consider the point with coordinates $(4, 8)$. How many units above the origin is this point located?

13 Consider the point with coordinates $(2, 7)$. How many units to the right of the origin is this point located?

14 Consider the point $A\,(4, 6)$. Find the coordinates of point A if it moves:

a 4 units to the right.

b 4 units to the left.

c 4 units upward.

d 4 units downward.

Distance between points

15 For each of the following, find the distance between the two points:

a $A\,(3, 6)$ and $B\,(8, 6)$.

b $C\,(10, 2)$ and $D\,(3, 2)$.

c $P\,(7, 5)$ and $Q\,(7, 9)$.

d $R\,(9, 12)$ and $S\,(9, 4)$.

16 Consider the points: $A\,(2, 9)$, $B\,(10, 9)$ and $C\,(10, 5)$

a Plot the points on a number plane.

b Find the length of AB.

c Find the length of BC.

17 A triangle has vertices with coordinates $A\,(5, 9)$, $B\,(1, 2)$ and $C\,(7, 2)$.

a Plot the triangle ABC on a number plane.

b Find the perpendicular height of the triangle if BC is the base.

c Find the length of the base BC.

18 A quadrilateral has vertices with coordinates $P\,(3, 8)$, $Q\,(6, 8)$, $R\,(1, 5)$ and $S\,(8, 5)$.

a Plot the quadrilateral $PQRS$ on a number plane.

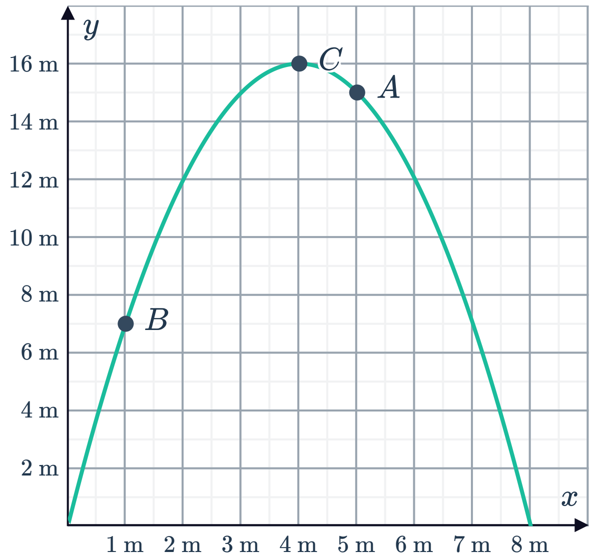
b Find the length of PQ.

c Find the length RS.

d State what type of quadrilateral $PQRS$ is.

Applications

19 A fire hose is turned on. The graph shows the height of the stream of water, in metres, in terms of its horizontal distance from the hose which is at the origin:



a Write down the coordinates of point A.

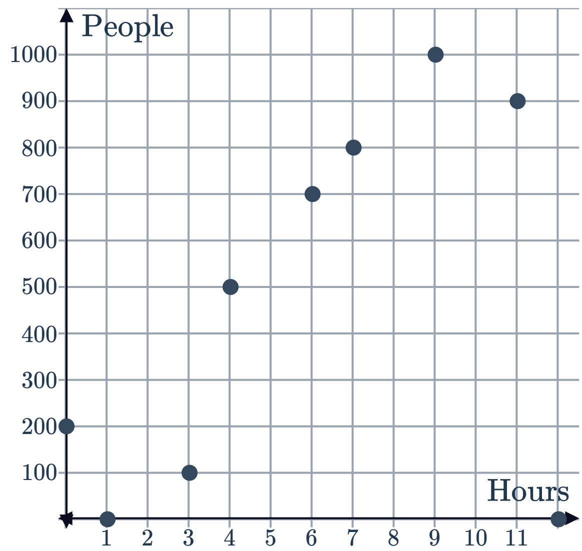
b Write down the coordinates of point B.

c Write down the coordinates of point C.

d The water reaches its maximum height at point C. Find the horizontal distance of point C from the hose.

e What is the maximum height reached by the water?

20 The number of people that logged onto a new website over a 12-hour period was recorded every hour as shown by the following graph:



a How many people were logged on when the site initially went live?

b How many people were logged on 6 hours after the site went live?

c After how many hours was the greatest number of people on the site recorded?

d What was the greatest number of people recorded on the site?